

Binary Distillation

Experiment 

CHE381: Unit Operations Laboratory I

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Executive Summary

Distillation is a crucial separation method in the industry, as it can provide an essentially pure separation under the right conditions. Distillation columns can be used in series to separate components present in the feed stream. Distillation works through phase equilibrium, as the more volatile component in the stream will leave the column as a vapor in the distillate, while the other component(s) leave as the bottoms product in the liquid phase. Each distillation column has a unique number of stages that is dependent on the process. The efficiency of the column depends on how well the distillation column achieves phase separation at each plate.

This lab had two objectives which were to determine the distillation column's variation with boil-up rate of pressure drop and to achieve a distillation by keeping a constant reflux ratio while varying top and bottom compositions with time, respectively. These objectives were completed by removing samples from the column at the top and bottom at varying voltages and recording the time it took to obtain a fixed volume of sample along with the refractive index of the sample obtained, the pressure difference, and the temperatures T1 and T8.

The main results of this lab were not as expected. However, the boil-up rate vs. the power level in the boiler and the temperature of T1 and T8 did make sense as the boil-up rate increased significantly with the power level and the temperature of T1 remained constant for the majority of the experiment while T8 varied slightly. The ethanol composition in the distillate was found to be approximately 0.8% for the majority of the trials, and the bottoms' ethanol composition ranged between 0.1-0.3%. The average volatility of the two components were calculated to be between 1.5 and 1.8, which is relatively high. The column efficiency was determined to be 54% which is quite low. There were approximately 4 calculated plates from the experimental values obtained, but in reality, there were 8 plates in the column. The overall errors can be observed when comparing the experimentally generated McCabe-Thiele graph with a theoretical McCabe-Thiele graph for water and ethanol.

Overall, the results obtained from this experiment were not ideal. The results were precise to each other but not accurate. The column efficiency was lower than expected, there were more plates in the column than the results showed, and the ethanol composition stayed somewhat constant throughout the experiment.

Introduction

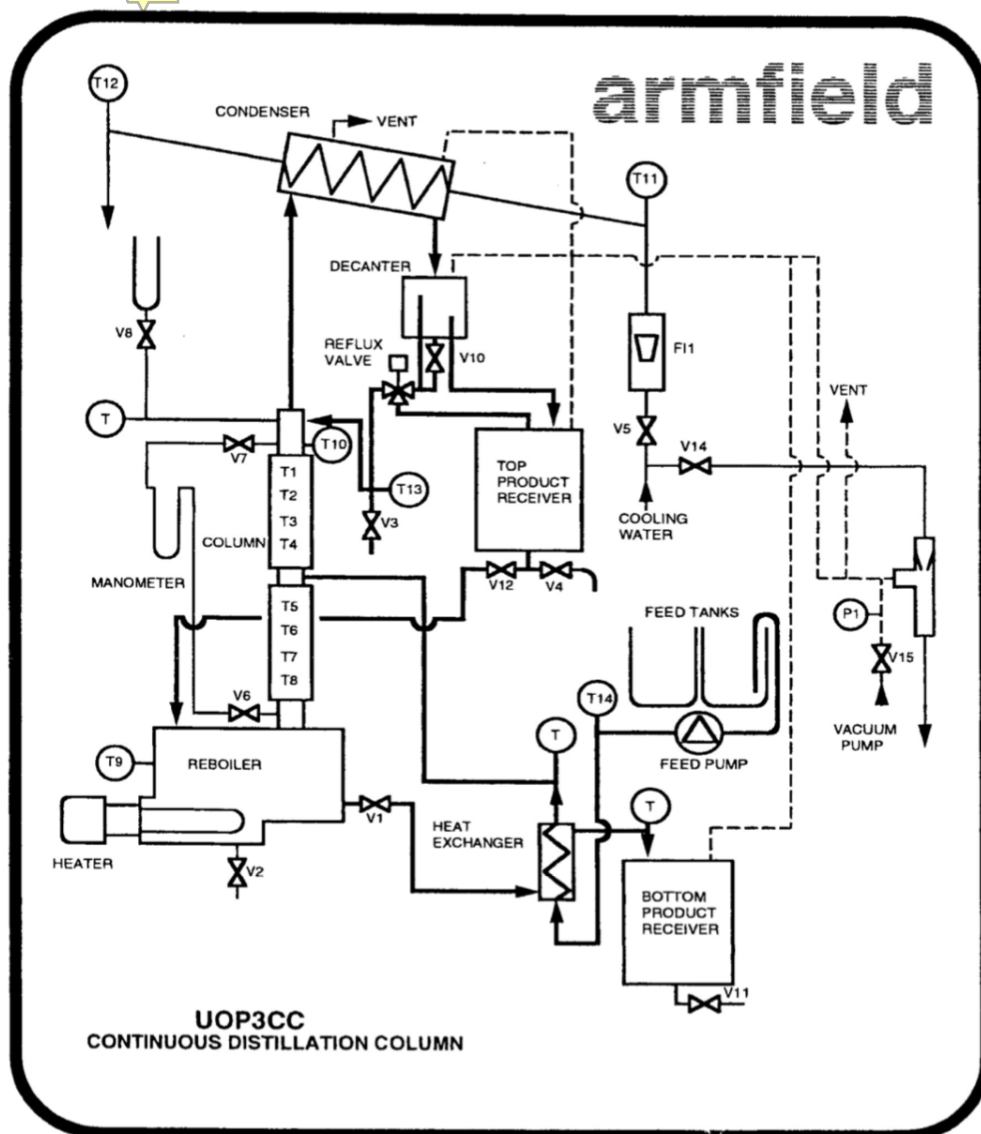
Distillation is an important unit operation that covers a range of applications, such as sea water desalination, separation of gasoline from crude oil, decaffeinating coffee beans, and purification of products in a downstream process. As a result of its versatility, distillation is widely used in the chemical and petroleum industry and accounts for 90-95% of all separation processes employed [1]. Typically, a mixture containing two or more non-miscible liquids that differ significantly in boiling points (at minimum 30°C) are most efficiently separated via distillation [2]. This makes it a suitable method for organic chemists to use this technique on a lab-scale to separate chemicals in the laboratory as well.

Being ubiquitous in all chemical industries, it is of great importance that one examines how distillation columns work and the surrounding parameters that affect its performance and efficiency. Hence, the objective of this experiment to determine the boil up rate and pressure drop over the distillation column, the column's overall efficiency, as well as observing the effects of the top and bottom compositions with varying reflux ratios.

For this experiment, a 10L 10wt% Ethanol-Water mixture will be used as the medium for separation in an Armfield UOP3CC batch distillation column unit (even though this apparatus a continuous distillation column by default, it will be configured as a batch column for this experiment). This lab scale column has eight internal trays which rely on equilibrium during operations. These trays are designed with caps or holes in which the vapor will flow through and the ones used in this experiment are called sieve trays. Weirs are also attached on the trays to trap liquid at a level just above the caps as gas condenses, which in turn allows for better separation. 

The ethanol-water mixture is poured into the reboiler, allowing it to heat up and vaporize its way past each contact stage which allows the component that has a lower boiling point to continue upwards until it condenses in the top receiver while the liquid with the higher boiling point will condense in the contact stage and flow back down into the bottoms receiver. A variable reflux drum is also installed at the top of the column to allow variation of the reflux ratio when determining the relationship, it has with respect to purity.

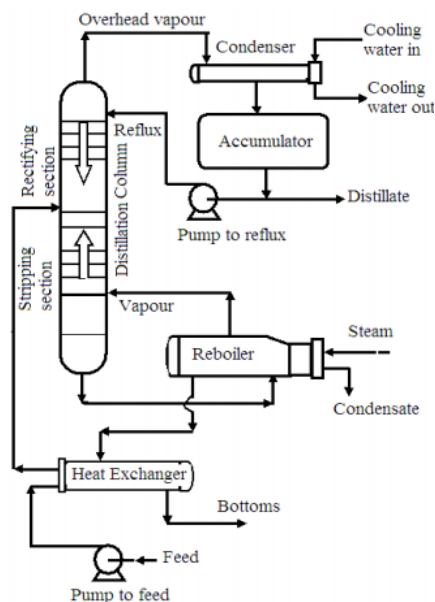
Fig. 1: Schematic diagram for the Armfield UOP3CC Unit



Theory

A typical continuous distillation is schematically presented in Figure 2. In most cases, a feed enters the middle of the column and is being separated where the component with a lower boiling point exits out at the top while the other component having the higher boiling point exits at the bottom.

Fig. 2: Schematic for continuous distillation process

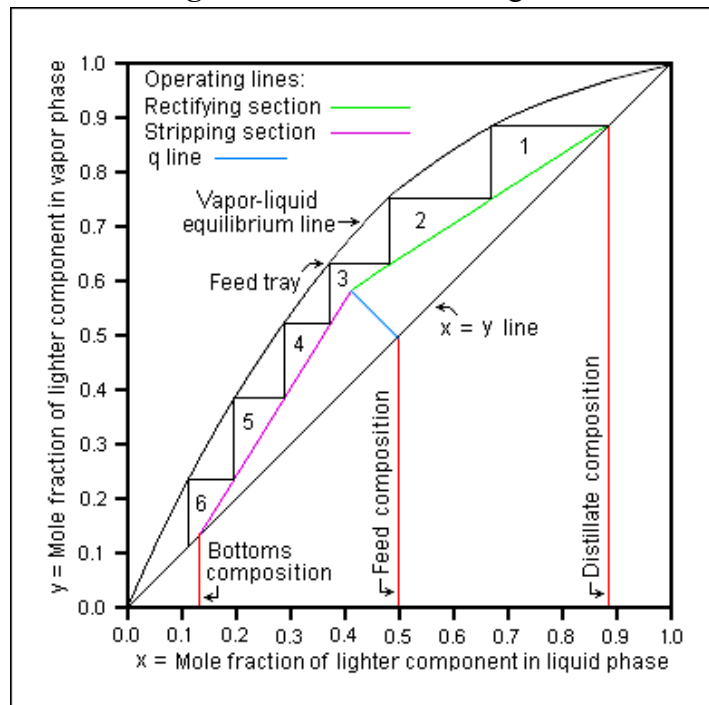


As this experiment is configured as a batch distillation process, the component with a higher boiling point will simply fall back into the reboiler instead of purging out of its own bottoms stream. An advantage to using batch distillation is that the system is easier to analyze and control. However, the equations pertaining to calculations are more complex especially when the system is comprised of more than two components.

Since this is a binary system, the McCabe-Thiele graphical technique can be used to model the distillation column and explore the effects of the various process variables such as number of theoretical stages require for the separation process.

Figure 3 depicts the components of a standard McCabe-Thiele plot. The graph is comprised of an equilibrium ($x=y$) line, a q line that represents the mole fraction of liquid, as well as 2 operating lines pertaining to the rectifying and stripping sections. In a batch process, there is only one operating line since the molar flow rates for both the vapor and liquid are constant throughout. A limitation with using the McCabe-Thiele plot lies with the assumptions made with regards to the operation of the distillation column. It assumes that every stage of the column obeys vapor-liquid equilibrium which is never the case in reality. Thus, it is necessary to calculate the efficiency of the column to make up for the shortfalls of the vapor-liquid equilibrium assumption.

Fig. 3: McCabe-Thiele Diagram



The following equation defines the efficiency of a column:

$$\eta = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad (1)$$

Where y_n^* represents the composition of vapor in equilibrium with liquid X_n . In the event that the vapor and liquid are both in thermodynamic equilibrium, y_n and y_n^* are both equal to each other.

Experiment 1: Determining the Effect of Boil-Up Rate and Overall Column Efficiency

For this experiment, it is important to consider the pressure drop during the operation of the distillation column. Pressure drops have a direct impact to the velocity of vapors traveling upwards and are directly related to each other. This means that an increased pressure drop will result in increased vapor velocity. ^[3] Pressure drops are a result of the condensate flowing back down the column via the outlet weir, which is also known as the down comer. This constant expansion and condensation of gas will in turn cause a change in pressure in the column.

The velocity of the vapor is being controlled by the power supplied to the reboiler. When liquid is absent, the pressure drop will be proportional to the square of the velocity. But due to the fact that there is liquid head, the proportionality relationship does not become applicable until foaming occurs.

Foaming is a factor that is affected by the amount of vapor passing through the liquids on the sieve trays and characterized by descriptors such as gentle or violent with regards to its behavior on each tray, and localized or spread out depending if it occurred on a fixed spot or throughout the tray.

Data collected for this run will include the boil up rate and pressure drop in cm.H₂O. The log/log relationship of each parameter will then be plotted on a graph and should theoretically show an increasing pressure drop with increased boil-up rates and more violent foaming.

To check if the number of actual plates was sufficient, Fenske's equation can be used to calculate the theoretical number of plates required for the separation process:

$$n + 1 = \frac{\log\left[\left(\frac{X_{EtOH}}{X_{H_2O}}\right)_d \left(\frac{X_{H_2O}}{X_{EtOH}}\right)_b\right]}{\log(\alpha_{EtOH/H_2O})_{avg}} \quad (2)$$

Where:

n: Number of theoretical plates

X_{EtOH} : Mole fraction of Ethanol

X_{H_2O} : Mole fraction of Water

$(\alpha_{EtOH/H_2O})_{avg}$: Average relative volatility

The subscripts b and d represent the bottoms and distillate respectively.

With the number of theoretical plates, the efficiency of the column can be calculated to determine if there were sufficient plates in the experimental model to efficiently separate the ethanol from water:

$$E = \frac{\text{Number of theoretical plates}}{\text{Number of actual plates}} \times 100\%$$

Experiment 2: Determining the Effect of Boil-Up Rate and Overall Column Efficiency

In this section of the experiment, the reflux ratio will be varied at 3:1 and 5:1 to determine what effect it has on compositions in both the top and bottom streams with respect to time. The data collected from this section will be the refractive index to determine the concentration and purity of the products. With that, the McCabe-Thiele plot can be plotted thereafter.

The material balance over the most volatile component follows the following equation:

$$V \times y_n = L \times x_{n+1} \quad (3)$$

Where V represents the vapor flow rate and L represents the liquid flow rate in the column respectively.

At total reflux, the flowrates of the vapor and liquid are equal. When $V=L$, equation (3) simplifies to:

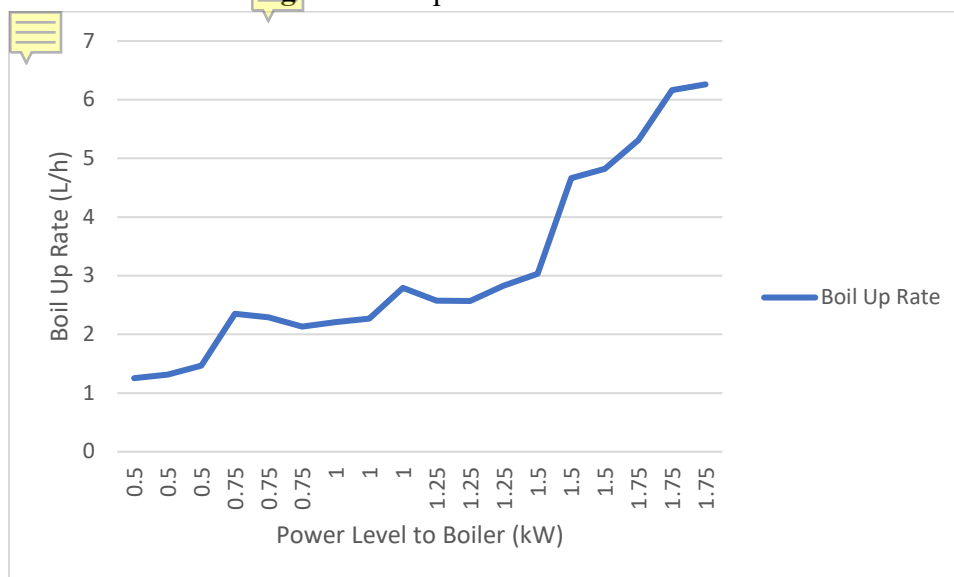
$$y_n = x_{n+1} \quad (4)$$

Results

The goal of this lab was to find the effects of boil up rate and reflux ratio on the experimental column efficiency. These two factors were measured into separate parts of the lab. A 10% weight percent solution of ethanol and water was used for the lab. For each part of the lab, samples of the distillate and bottoms were taken incrementally in order to measure the effects of changing boil up rate or reflux ratio. The boil up rate was varied by changing the amount of power supplied to the boiler. A total of 18 runs of data were collected for part A, 6 different power levels and 3 trials for each one. For part B, a total of 10 runs were collected on 2 different reflux ratios with 5 trials for both. The power level for part B was kept at a midpoint of 0.75 kW for gentle foaming to be observed.

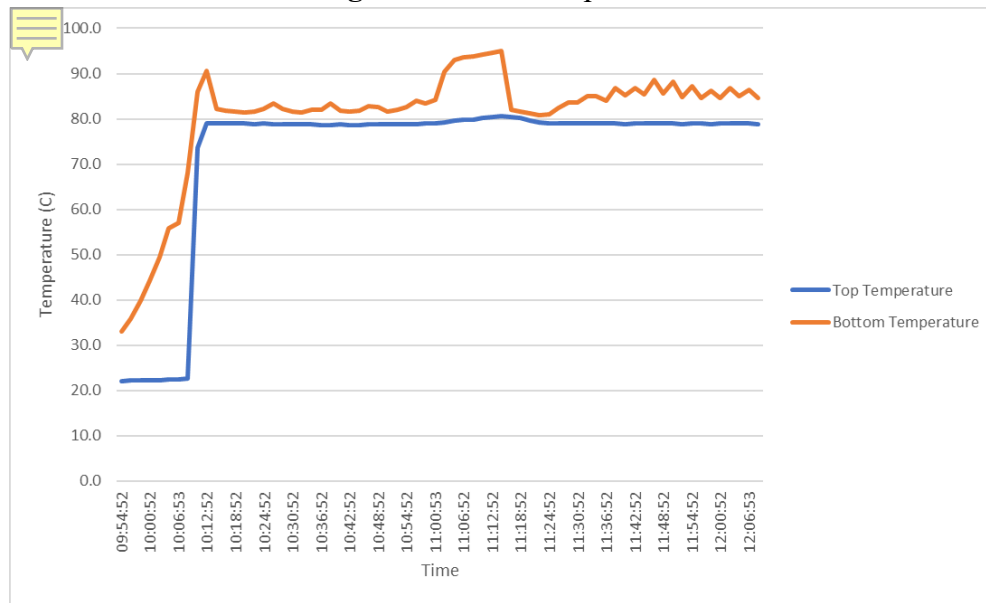
The results recorded in part A consisted of the boil up rate, pressure drop across the column, degree of foaming, column temperature and the refractive index of the samples collected. The power level to the boiler was taken starting at 0.5 kW up until 1.75 kW at 0.25 kW increments. It can be seen in Chart X that the boil up rate increases rapidly with the increase in power level.

Fig. 4: Boil Up Rate to Boiler Power



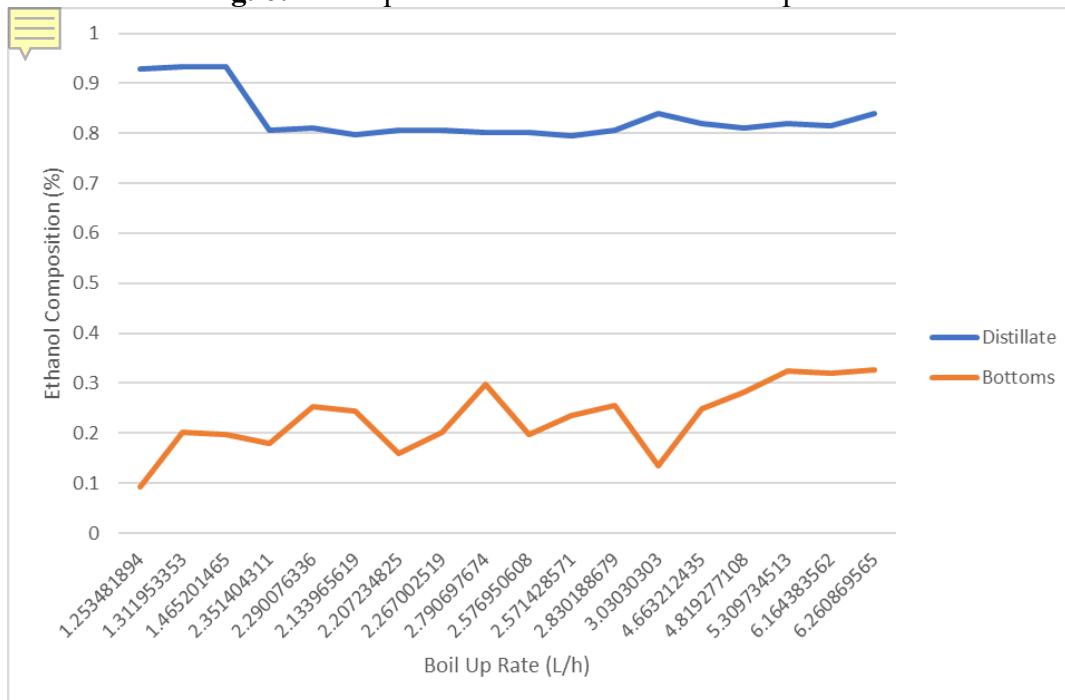
Temperature was incrementally recorded once the boiler had first initially heated up to around 70°C. Chart X shows the temperature of the top and bottom of the column relative to time, this shows how the column progressed in temperature over the course of the experiment.

Fig. 5: Column Temperature



The refractive index of each sample, both distillate and bottoms, was taken in order to find the composition of the ethanol by weight in the sample. The relation of refractive index to ethanol weight percent that was found when calibrating the refractometer was used here. Figure 4 shows the relation of boil up rate to ethanol weight percent.

Fig. 6: Boil Up Rate Effects on Ethanol Composition



It can be seen in Figure 6 that the distillate was relatively constant in ethanol composition, hovering around 80%. Initially the distillate showed a high ethanol percentage but then stayed constant for the rest of the experiment. As for the bottom product, the ethanol composition stayed totally constant at around 25% over the course of the experiment.

Theoretically, Ethanol should be the most volatile compound and the water is less so. The experimental data shows the same, as seen in Table 1.

Table 1: Volatility of Components

Relative Volatility Ethanol	Relative Volatility Water	Volatility AB
14.04	0.31	2.08
26.38	0.22	2.39

Since the difference in volatility for the two components is considerably large, the average volatility of both components is found to be between 1.5 and 1.8. This is rather high for most systems and most likely explains why the composition values seemed to stay constant. Both the measured volatility and the composition values influence the measured effectiveness of the column. The measured effectiveness is determined on the number of theoretical plates compared to the number there actually were. In the real system there was a total of eight plates, including a reboiler. The theoretical plates that were calculated can be seen below in Table 2.

Table 2: Boil Up Rate Effects on Ethanol Wt% and Theoretical Plates

Boil up rate (L/h)	Wt% Ethanol	Theoretical Plates
1.253481894	0.928333494	7.020622539
1.311953353	0.932534098	
1.465201465	0.932534098	
2.351404311	0.806515977	3.500239261
2.290076336	0.810716581	
2.133965619	0.798114769	
2.207234825	0.806515977	3.469126411
2.267002519	0.806515977	
2.790697674	0.802315373	
2.576950608	0.802315373	3.475764589
2.571428571	0.793914165	
2.830188679	0.806515977	
3.03030303	0.840120809	3.7635821
4.663212435	0.819117789	
4.819277108	0.810716581	
5.309734513	0.819117789	4.960306521
6.164383562	0.814917185	
6.260869565	0.840120809	

Table 3 shows that the theoretical plates follow the same trend as the ethanol composition in the top stream, where the value for both is initially high and then drops to a constant value. Adding on the efficiency to Table 3 shows how exactly varying the boil up rate had affected the efficiency of the column.

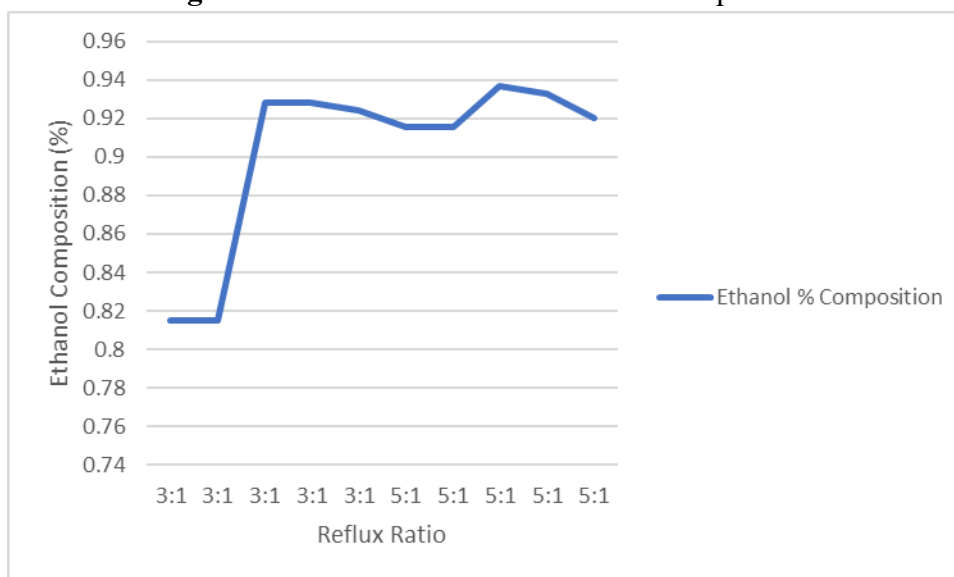
Table 3: Boil Up Rate effects on Theoretical Plates and Efficiency

Boil up rate (L/h)	Wt% Ethanol	Theoretical Plates	Efficiency (%)
1.253481894	0.928333494	7.020622539	87.75778174
1.311953353	0.932534098		
1.465201465	0.932534098		
2.351404311	0.806515977	3.500239261	43.75299076
2.290076336	0.810716581		
2.133965619	0.798114769		
2.207234825	0.806515977	3.469126411	43.36408014
2.267002519	0.806515977		
2.790697674	0.802315373		
2.576950608	0.802315373	3.475764589	43.44705737
2.571428571	0.793914165		
2.830188679	0.806515977		
3.03030303	0.840120809	3.7635821	47.04477625
4.663212435	0.819117789		
4.819277108	0.810716581		
5.309734513	0.819117789	4.960306521	62.00383152
6.164383562	0.814917185		
6.260869565	0.840120809		

Table 3 shows that boil up rate had no real effect on the efficiency of the column since everything stayed very constant. Overall, the column was found to be average of 54% efficiency. The results show that experimentally, the column only had 4 plates when in reality the column had 8 plates.

Part B had a similar goal of finding the column effectiveness but this time the boil up rate was kept constant while the reflux ratio was varied. Two reflux ratios were test, 3:1 and 5:1, were a set amount was sent back to the column and then a smaller portion was sent to be collected. For this part both the top and bottom samples were taken at the same time in order for the reflux ratio to not have much an effect on the data. Figure 7 shows a similar result as what was observed in part A. The composition of ethanol stayed constant for both distillate and bottoms samples.

Fig. 7: Reflux Ratio effect on Ethanol Composition



For the distillate, ethanol started off as a bit lower in composition initially but then proceeded to stay almost constant throughout the rest of the experiment. Figure 7 also shows that the reflux ratio did not have any effect on the composition of ethanol in the samples since they stayed constant. The volatility of the compounds did not change in between parts A and B. Following the same calculation process as in part A, Table 4 shows the effect of reflux ratio on the number of theoretical plates.

Table 4: Reflux Ratio Effects on Theoretical Plates

Reflux Ratio	Theoretical Plates
3:1	4.230591444
5:1	4.517951487

The number of plates in Table 5 are based of averages of 5 runs for both reflux ratios that were tested. The number of plates only increased by 0.25 which is barely any change when the reflux ratio was changed by a large amount. Using these plate values, the column efficiency can be seen below in table 5.

Table 5: Reflux Ratio Effects on Efficiency Plate

Reflux Ratio	Efficiency (%)
3:1	52.88239306
5:1	56.47439358

Just as in part A, the column efficiency was measure out to be 54%. This result is consistent throughout the lab, but neither are very accurate to the actual value.

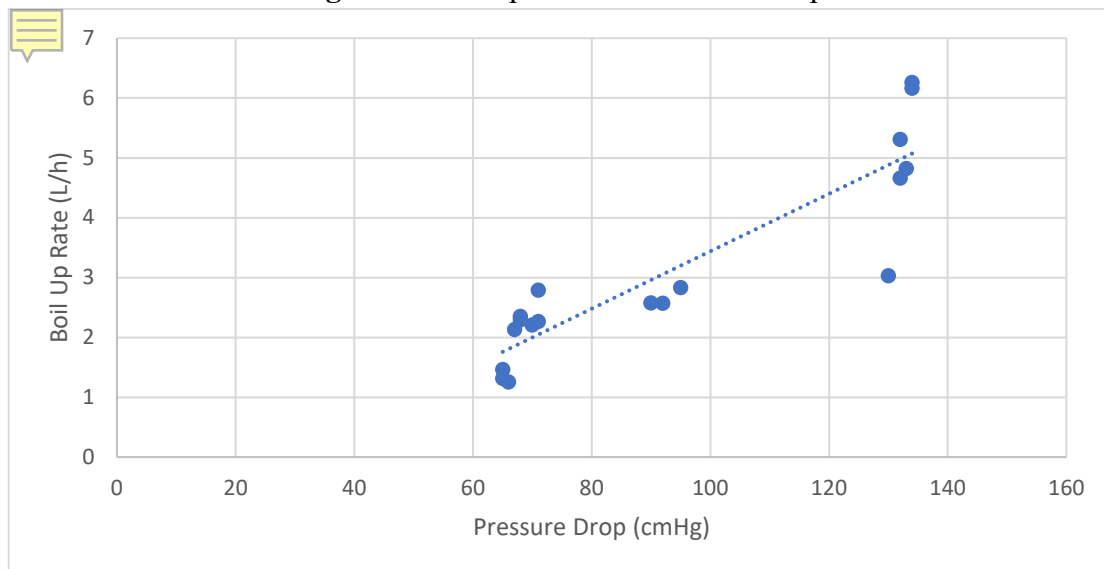
Discussion

Distillation columns are a great way to separate liquids of two different boiling points. The ideal system would operate at steady state. Steady state entails that the quantities of solution do not change over time. Some of the quantities that should not change to be considered at steady state include flow rate of the feed, flow rate of the outputs, the heating and cooling rates, the reflux ratio, the pressures within the system, the compositions at one location, and the temperatures.

The column that is being worked with in the system had not reached a steady state due to the temperature changing, especially at the top of the column. If the column was run for a longer time the system may have come to steady state but within the given time constraints the column was not at steady state. This can also be seen by the pressure change not remaining consistent.

The relationship between boil up rate and pressure is a direct relationship. As pressure drop increases so does the boil up rate. The relation can be shown in the graph shown below.

Figure 8: Boil up Rate vs. Pressure Drop

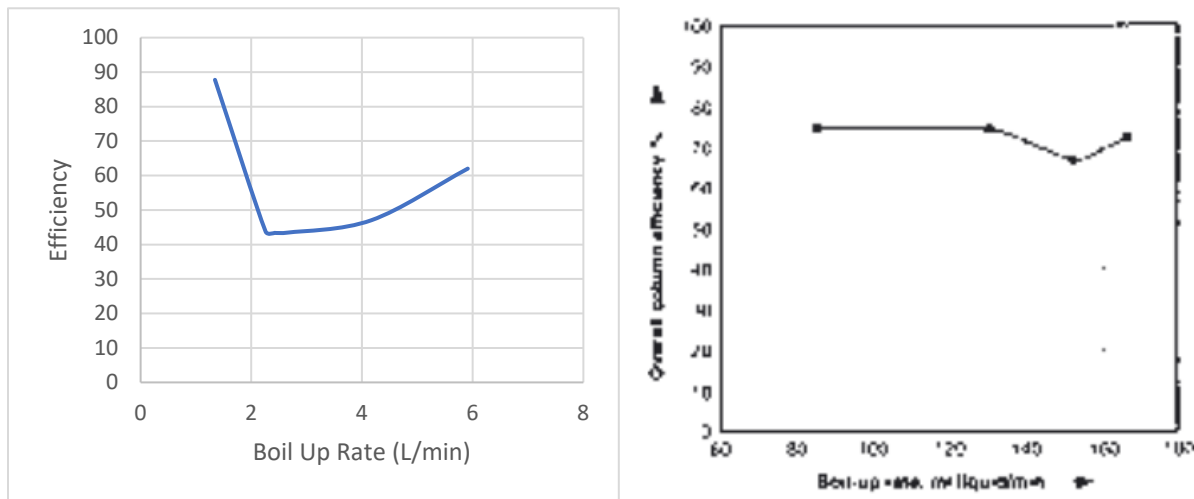


This relationship makes sense due to the fact that when the boil up rate increases the amount of vapor within the system also increases. The vapor is trying to travel up the column while the liquid is traveling back down the column. The vapor flowing up the column creates resistance within the system which in turn increases the pressure drop in the column.

The relationship between boil-up rate and column efficiency is not very clear when looking at the data obtained in the experiment.

Figure 9: Experimental Boil-Up Rate vs. Column Efficiency

Figure 10: Theoretical Boil-Up Rate vs. Column Efficiency



Comparing to results found in literature, the relationship between boil up rate and efficiency is that there isn't one. The efficiency remains fairly constant after the first boil-up rate was measured. This was consistent with figure 10 obtained from literature.

The efficiency of the column in the lab is far less than those used in industry. The distillation columns used within industry are much larger than the one used in lab. The equipment that is being used within industry is also more advanced and better calibrated. The distillation columns within industry regulate the pressure better, the temperature better, and the flow rates within the setup.

The errors within the system can definitely be seen when observing the McCabe-Thiele graph produced with the data obtained in lab. The results that were obtained in lab did not correlate with the McCabe-Thiele diagram presented in the theory section due to the number of moles of ethanol in the solution that was obtained being fairly constant from measurement to measurement. The consistency in numbers can be seen when at a power of 0.75 kW, 1 kW, and 1.25 kW. The consistency and slight variation of moles did not allow for the increase in mole fraction of ethanol in the vapor and in the liquid simultaneously.

Throughout the experiment there were several assumptions that were made. One assumption that was made in the experiment is that the reboiler and condenser had the same load. It was also assumed that the only source of heating and cooling was from the condenser and reboiler.[1] Finally, it was assumed that the trays were being 100% efficient. All of these assumptions are not accurate assumptions for this experiment.

One of the reasons that the assumptions of this experiment are incorrect is because assuming that the loads of the condenser and the reboiler are equal is incorrect due to the fact that there is often heat carried in by the feed stream or lost when hot liquid is removed from the system. If the load difference between the condenser and reboiler is small then it can be ignored, but if it is large the hot and cold utility must be considered and a plot must be created to find pinch point. Further calculations would be done after to account for heat being both added and lost.

When assuming the efficiency is at 100% it is thought that the trays are operating at their maximum capability. This also means that the vapor and liquid phases are assumed to be at equilibrium. This also assumes that all the trays have the same performance. [2] However, not all the trays will have the same efficiency. This is because the amount of liquid and vapor phase is different throughout the whole system which changes the efficiency of each tray in the column.

Due to the amount of incorrect assumptions, this lead to different results than expected. However, there was also equipment and human error involved in the difference in data. One of the largest sources of error was when using the refractometer. Between each measurement, the refractometer must be zeroed by using water and then cleaning it with chem paper. Not cleaning the lens well enough could give results that are inaccurate.

Overall, the experiment did not obtain the expected results, especially when looking at the McCabe-Thiele graph. The distillation column was able to do its job by separating ethanol and water, which was seen from the vapor accumulating in the column. However, error lead us to results that did not agree with theoretical values.

Resources

1. CKemp, Ian. "Process Change and Evolution." *ScienceDirect*, Butterworth-Heinemann, 20 May 2008, www.sciencedirect.com/science/article/pii/B9780750682602500110.
2. "DISTILLATION TRAY EFFICIENCY." *Neutrium*, May 2018, neutrium.net/equipment/distillation-tray-efficiency/.

Appendix I: Data Tabulation/Graphs

Fig. 4: Boil Up Rate to Boiler Power

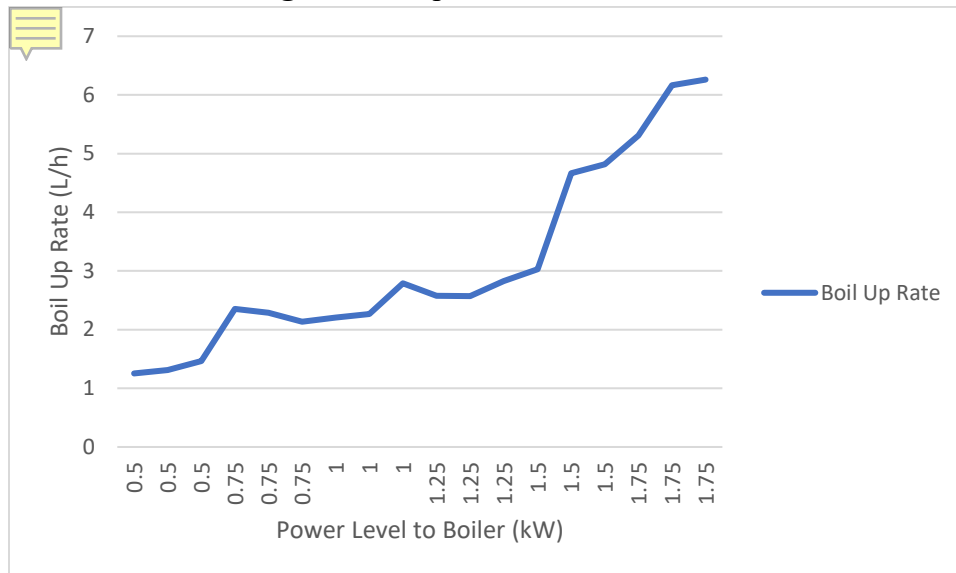


Fig. 5: Column Temperature

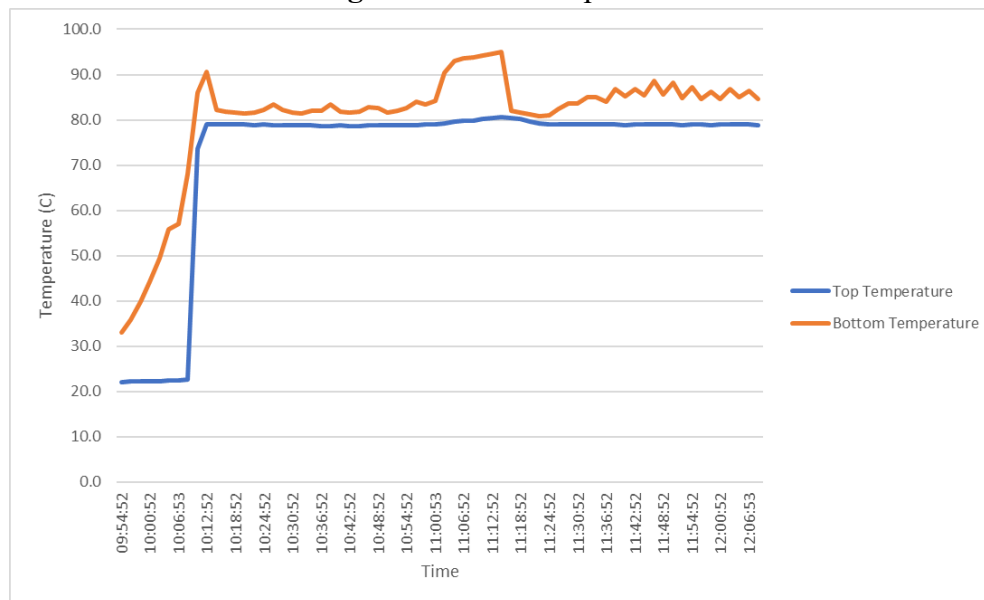


Fig. 6: Boil Up Rate Effects on Ethanol Composition

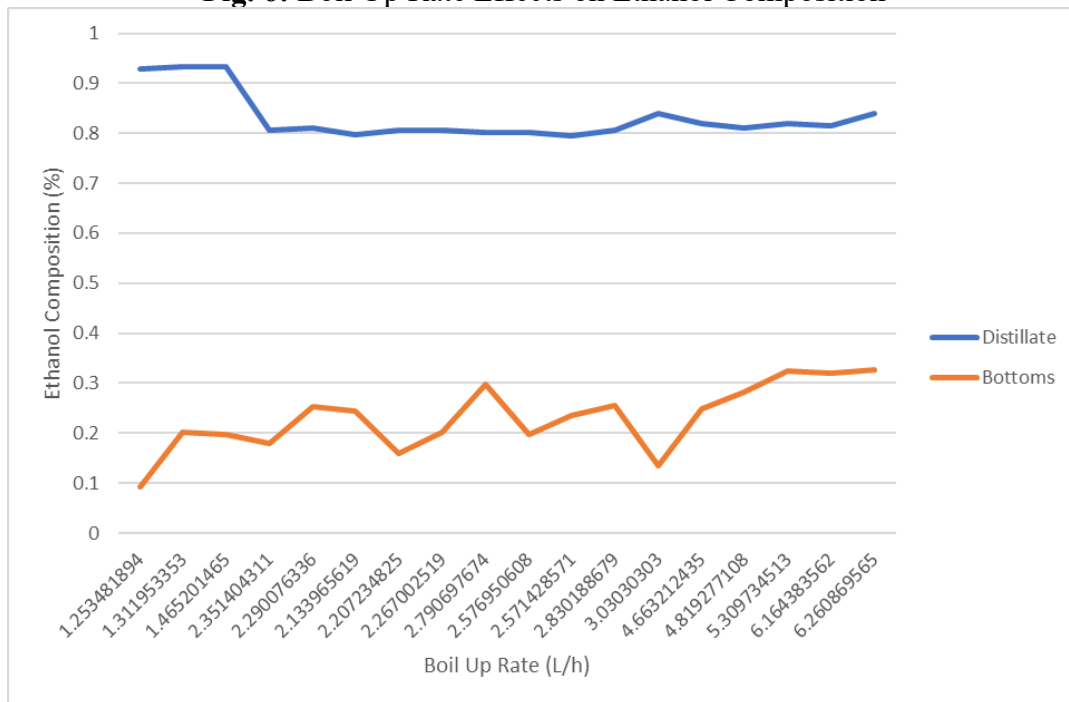


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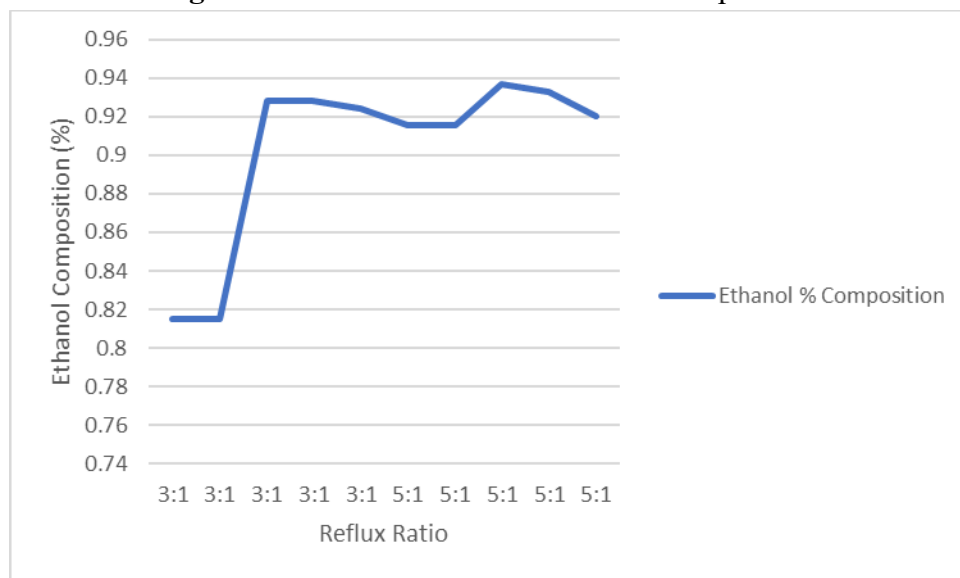


Figure 8: Boil up Rate vs. Pressure Drop

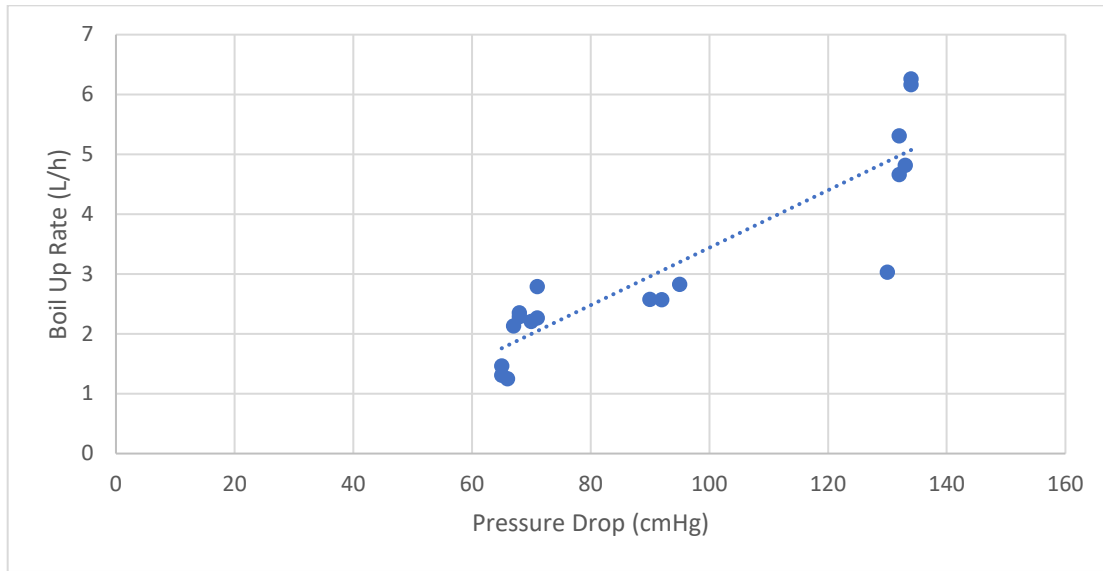


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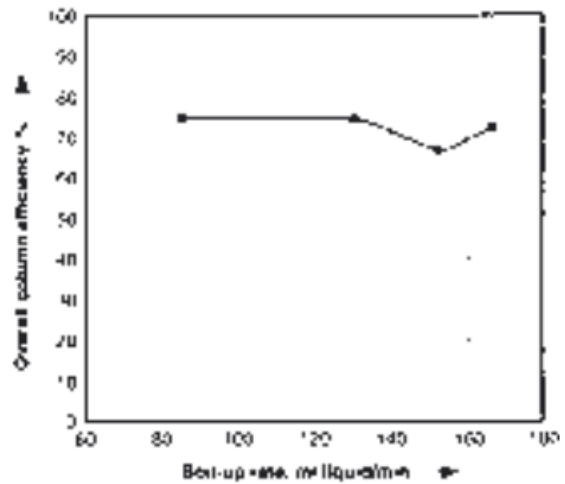
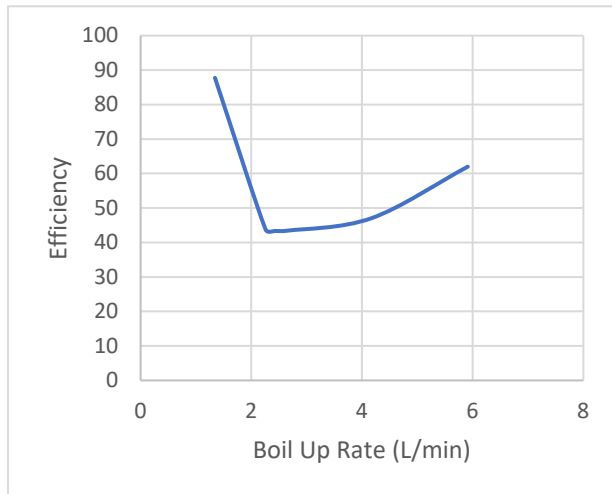


Figure 11: Refraction vs. wt% Ethanol

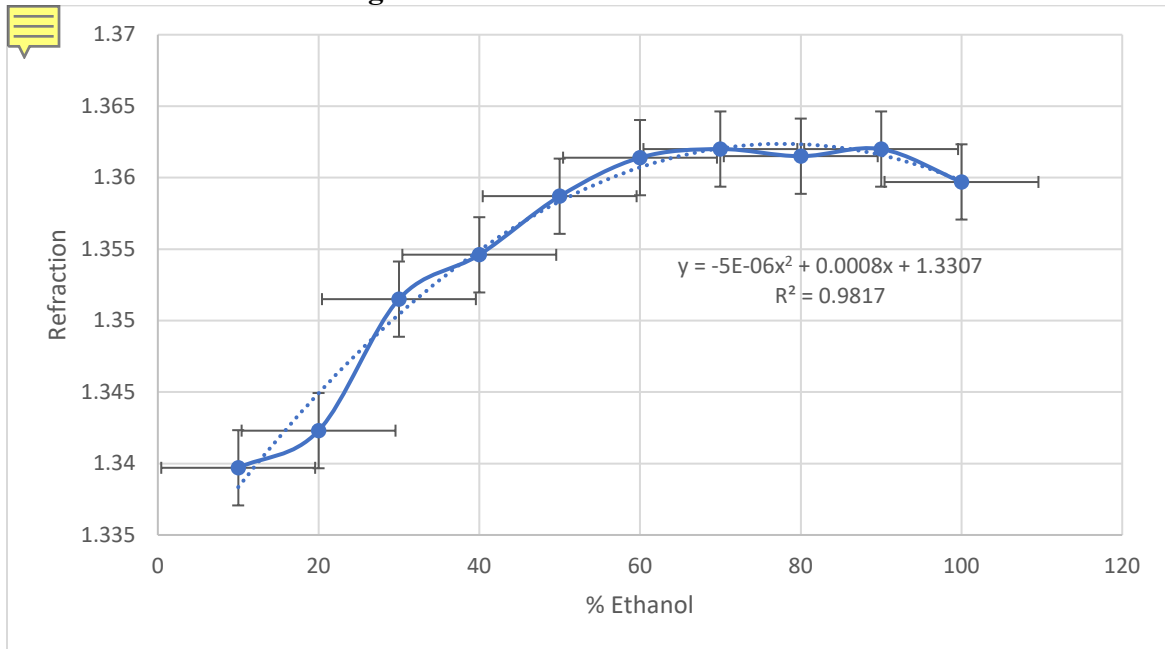


Figure 12: McCabe-Thiele Diagram drawn from Experimental Results

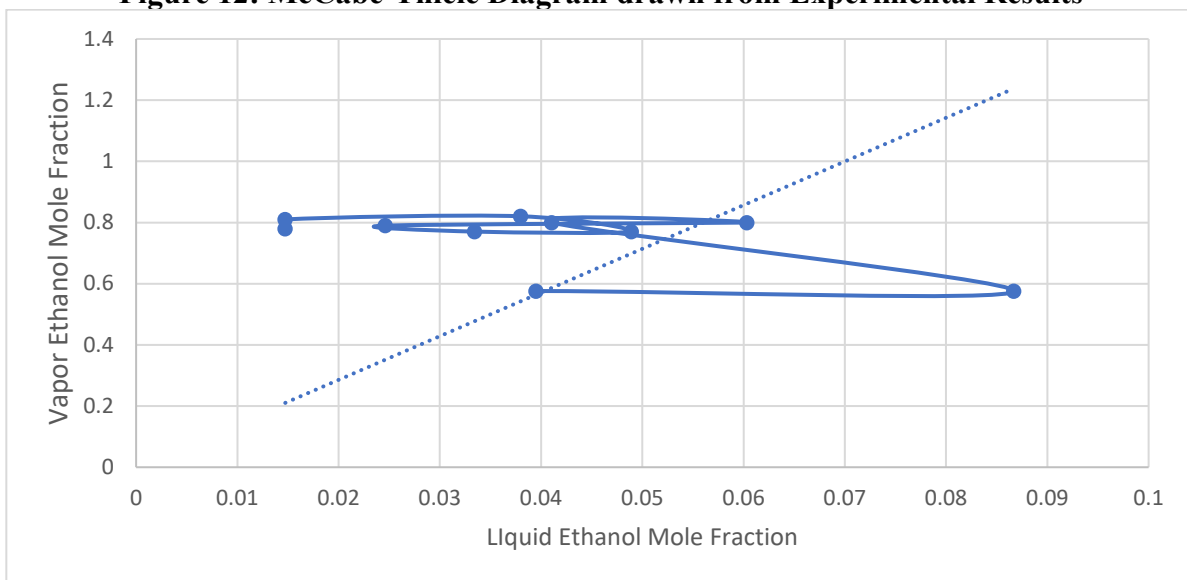


Figure 13: Temperature Vs Time within the Column

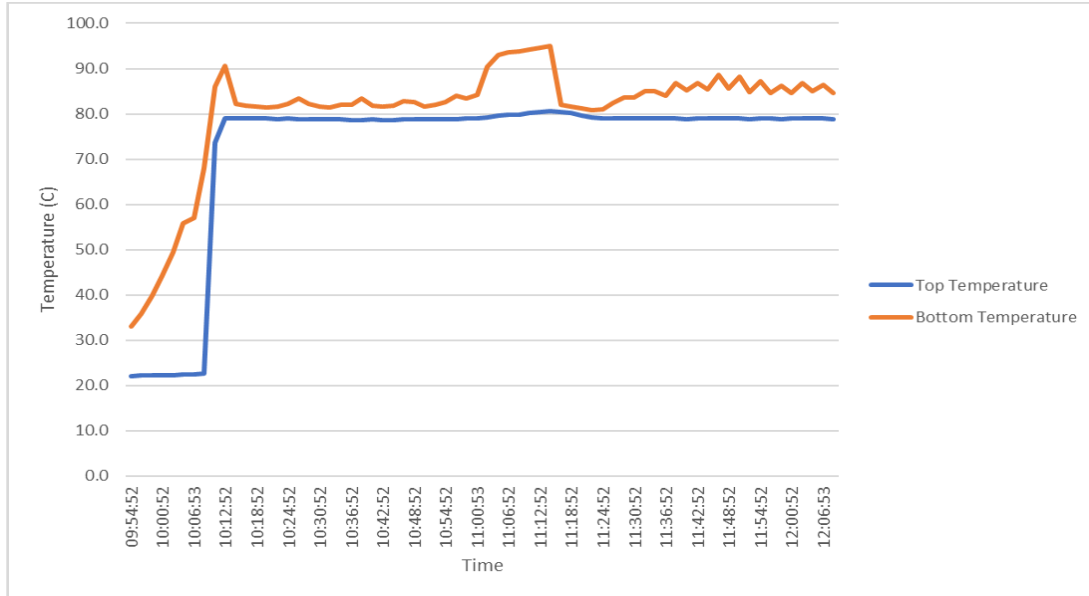


Figure 14: Boil Up Rate Vs Ethanol Composition

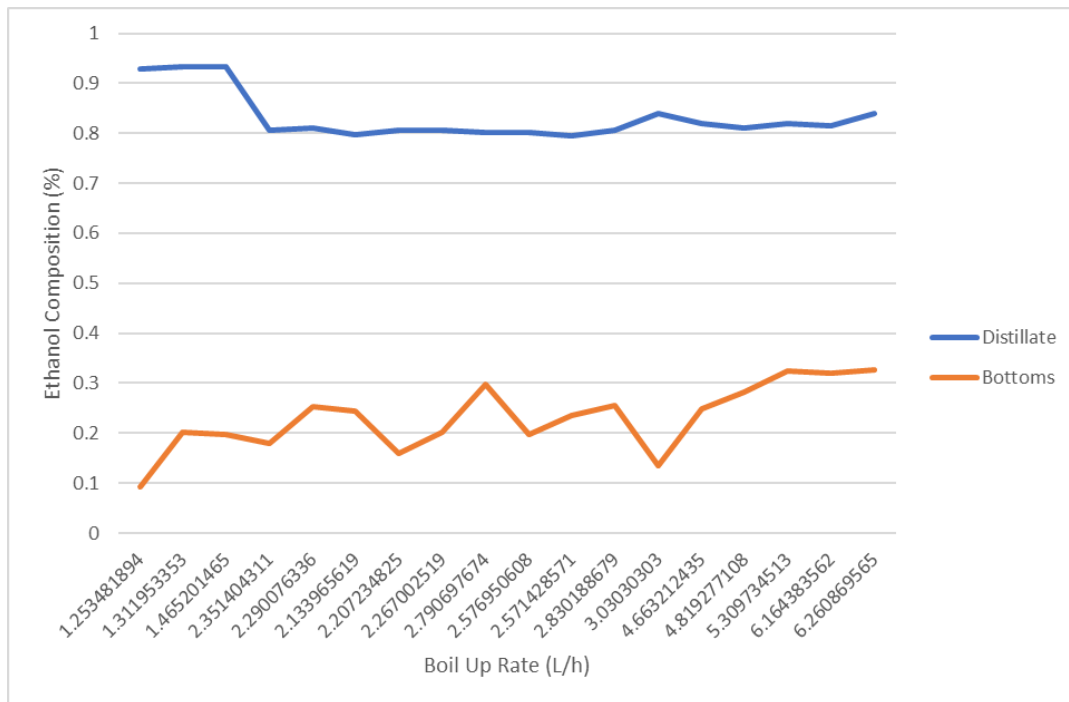


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2.571428571	0.793914165		
2.830188679	0.806515977		
3.03030303	0.840120809	3.7635821	47.04477625
4.663212435	0.819117789		
4.819277108	0.810716581		
5.309734513	0.819117789	4.960306521	62.00383152
6.164383562	0.814917185		
6.260869565	0.840120809		

Table 4: Reflux Ratio Effects on Theoretical Plates

Reflux Ratio	Theoretical Plates
3:1	4.230591444
5:1	4.517951487

Table 5: Reflux Ratio Effects on Efficiency Plate

Reflux Ratio	Efficiency (%)
3:1	52.88239306
5:1	56.47439358

Table 6: Refractometer Calibration Data

% ethanol	Refraction Values			Average	Standard Deviation	RSD
10	1.3397	1.3394	1.3396	1.339566667	0.000152753	0.01140313
20	1.3423	1.3421	1.3423	1.342233333	0.00011547	0.00860283
30	1.3515	1.3513	1.3514	1.3514	1E-04	0.007399734
40	1.3546	1.3546	1.3547	1.354633333	5.7735E-05	0.004262041
50	1.3587	1.3586	1.3586	1.358633333	5.7735E-05	0.004249493
60	1.3614	1.3616	1.3613	1.361433333	0.000152753	0.011219978
70	1.362	1.362	1.362	1.362	0	0
80	1.3615	1.3614	1.3616	1.3615	1E-04	0.00734484
90	1.362	1.36	1.364	1.362	0.002	0.146842878
100	1.3597	1.3599	1.3598	1.3598	0.0001	0.007354023

Table 7: Top Boil-Up Rate Results for Part 1

Power (kW)	Boil up rate (L/h)	Pressure Drop (cmH₂O)	Degree of Foaming on Trays
0.5	1.253481894	66	Gentle, localized
0.5	1.311953353	65	Gentle, localized
0.5	1.465201465	65	Gentle, localized
0.75	2.351404311	68	Gentle, localized
0.75	2.290076336	68	Gentle, localized
0.75	2.133965619	67	Gentle, localized
1	2.207234825	70	Rigorous, spread out
1	2.267002519	71	Rigorous, spread out
1	2.790697674	71	Rigorous, spread out
1.25	2.576950608	90	Vigorous, spread out
1.25	2.571428571	92	Vigorous, spread out
1.25	2.830188679	95	Vigorous, spread out
1.5	3.03030303	130	Top is vigorous, bottom is gentle and even, both spread out
1.5	4.663212435	132	Top is vigorous, bottom is gentle and even, both spread out
1.5	4.819277108	133	Top is vigorous, bottom is gentle and even, both spread out
1.75	5.309734513	132	Top is vigorous, bottom is gentle and even, both spread out
1.75	6.164383562	134	Top is vigorous, bottom is gentle and even, both spread out
1.75	6.260869565	134	Top is vigorous, bottom is gentle and even, both spread out

Table 8: Top Boil-Up Rate Results for Part 2

Power (kW)	Boil up rate (L/h)	Average Pressure Drop (cmH₂O)	Degree of Foaming on Trays
0.75	2.07253886	72	Gentle, Localized
0.75	2.211302211	74	Gentle, Localized
0.75	1.949106659	70	Gentle, Localized
0.75	1.95545899	74	Gentle, Localized
0.75	2.021336328	75	Gentle, Localized
0.75	1.979109401	71	Gentle, Localized
0.75	2.66075388	74	Gentle, Localized
0.75	2.56593015	75	Gentle, Localized
0.75	2.567760342	75	Gentle, Localized
0.75	2.267002519	76	Gentle, Localized

Table 9: Weight percent and mole fractions for Problem 1

	Top			Bottom		
Power	Wt% E	mol frac E Ya	mol frac W Yb	Wt% E	mol frac E Xa	mol frac W Xb
0.5	0.92833	0.79990	0.20010	0.09241	0.03047	0.96953
0.5	0.93253	0.81009	0.18991	0.20163	0.07230	0.92770
0.5	0.93253	0.81009	0.18991	0.19743	0.07056	0.92944
0.75	0.80652	0.56263	0.43737	0.18063	0.06370	0.93630
0.75	0.81072	0.56930	0.43070	0.25204	0.09420	0.90580
0.75	0.79811	0.54956	0.45044	0.24364	0.09042	0.90958
1	0.80652	0.56263	0.43737	0.15962	0.05537	0.94463
1	0.80652	0.56263	0.43737	0.20163	0.07230	0.92770
1	0.80232	0.55605	0.44395	0.29824	0.11595	0.88405
1.25	0.80232	0.55605	0.44395	0.19743	0.07056	0.92944
1.25	0.79391	0.54315	0.45685	0.23523	0.08670	0.91330
1.25	0.80652	0.56263	0.43737	0.25624	0.09610	0.90390
1.5	0.84012	0.61856	0.38144	0.13442	0.04573	0.95427
1.5	0.81912	0.58291	0.41709	0.24784	0.09230	0.90770
1.5	0.81072	0.56930	0.43070	0.28144	0.10784	0.89216
1.75	0.81912	0.58291	0.41709	0.32345	0.12857	0.87143
1.75	0.81492	0.57606	0.42394	0.31925	0.12643	0.87357
1.75	0.84012	0.61856	0.38144	0.32765	0.13073	0.86927

Table 10: Weight percent and mole fractions for Problem 2

	Top			Bottom		
Power (kW)	Wt% E	mol frac E Ya	mol frac W Yb	Wt% E	mol frac E Xa	mol frac W Xb
0.75	0.814917	0.576059	0.423941	0.117617	0.039511	0.960489
0.75	0.814917	0.576059	0.423941	0.235234	0.086696	0.913304
0.75	0.928333	0.799905	0.200095	0.121818	0.041052	0.958948
0.75	0.928333	0.799905	0.200095	0.172225	0.060335	0.939665
0.75	0.924133	0.789880	0.210120	0.075611	0.024622	0.975378
0.75	0.915732	0.770308	0.229692	0.100814	0.033444	0.966556
0.75	0.915732	0.770308	0.229692	0.142821	0.048905	0.951095
0.75	0.936735	0.820449	0.179551	0.113416	0.037980	0.962020
0.75	0.932534	0.810093	0.189907	0.046207	0.014731	0.985269
0.75	0.919932	0.780016	0.219984	0.046207	0.014731	0.985269

Table 11: Problem 1 Data

Power (kW)	Volatility AB	Theoretical Plates	Efficiency
0.5	1.6925	7.0206	87.7578
0.75	1.8013	3.5002	43.7530
1	1.8170	3.4691	43.3641
1.25	1.7876	3.4758	43.4471
1.5	1.7929	3.7636	47.0448
1.75	1.4679	4.9603	62.0038

Table 12: Problem 2 Data

Reflux Ratio	Volatility AB	Theoretical Plates	Efficiency
0.1257	2.0768	4.2306	52.8824
0.2090	2.3883	4.5180	56.4744

Appendix II: Error Analysis

Error occurred throughout the entirety of the lab. Most of the error that occurred in the lab was due to human error and timing or measurement difficulties. There was also error that occurred with the equipment, especially the refractometer. The initial calibration curve was expected to be a parabola. The values on the calibration curve had a general shape of a parabola but the values did not come down as much as expected when the weight percent of ethanol was increased. This leads to error in calculation, especially when calculating the weight percent of ethanol. The weight percent of ethanol was then used throughout the rest of the calculations which shows how important the calibration curve is. This error could have occurred because of the refractometer not being cleaned properly or from not putting enough solution on the refractometer.

Throughout the lab there were times when collecting ethanol that the solution would splash and not make it into the beaker. This would affect the collection time and has a chance of effecting the refraction values. Also, when collecting there was an inconsistency in flow coming out of the valves. A large amount would come out at first and then would continue to slowly drip until the valve was closed. This would affect the time recorded to collect 10 mL of solution.

Table 6: Refractometer Calibration

% ethanol	Refraction Values			Average	Standard Deviation	RSD
10	1.3397	1.3394	1.3396	1.339566667	0.000152753	0.01140313
20	1.3423	1.3421	1.3423	1.342233333	0.00011547	0.00860283
30	1.3515	1.3513	1.3514	1.3514	1E-04	0.007399734
40	1.3546	1.3546	1.3547	1.354633333	5.7735E-05	0.004262041
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60	1.3614	1.3616	1.3613	1.361433333	0.000152753	0.011219978
70	1.362	1.362	1.362	1.362	0	0
80	1.3615	1.3614	1.3616	1.3615	1E-04	0.00734484
90	1.362	1.36	1.364	1.362	0.002	0.146842878
100	1.3597	1.3599	1.3598	1.3598	0.0001	0.007354023

As shown in the table above, the standard deviation is relatively small. This means that all of the values are close to each other when being measured at a specific ethanol percentage. However, just because the values are precise, doesn't mean that they are accurate. The values are close to each other, but this does not mean that the values are what they are supposed to be. All the measurements could be wrong. The only way to know is comparing to a theoretical value and calculating the percent error between the data points.



Appendix III: Sample Calculations

$$\text{Wt \% Ethanol} = \frac{\text{Refractive index} - \text{Calibrated Refractive index}}{\text{Slope of Refraction vs wt \% graph}} = \frac{1.3643 - 1.3422}{.0238061} = .9283 \text{ wt \% Ethanol}$$

$$\text{Average Wt \% Ethanol} = \frac{\sum \text{wt \% ethanol at a given power}}{n} = \frac{.9283 + .9325 + .9325}{3} = .9311 \text{ wt \% Ethanol}$$

$$\text{Moles Ethanol} = \frac{\text{wt \% Ethanol} * \text{Volume} * \text{Density}}{\text{Molar Mass Ethanol}} = \frac{.92833 * 10 \text{ mL} * .789 \frac{\text{g}}{\text{mL}}}{46.07 \frac{\text{g}}{\text{mol}}} = .1590 \text{ moles Ethanol}$$

$$\text{Moles Water} = \frac{(1 - \text{wt \% Ethanol}) * \text{Volume}}{\text{Molar Mass Water}} = \frac{(1 - .92833) * 10 \text{ mL} * 1 \frac{\text{g}}{\text{mL}}}{18.02 \frac{\text{g}}{\text{mol}}} = .03977 \text{ moles Water}$$

$$\text{Mole Fraction of Ethanol } (Y_a) = \frac{\text{moles Ethanol}}{\text{moles Ethanol} + \text{moles Water}} = \frac{.1590}{.1590 + .03977} = .7999$$

$$\text{Mole Fraction of Water } (Y_b) = \frac{\text{moles Water}}{\text{moles Ethanol} + \text{moles Water}} = \frac{.03977}{.1590 + .03977} = .2001$$

$$\text{Average } Y_a = \frac{\sum Y_a \text{ at a given power}}{n} = \frac{.7999 + .8101 + .8101}{3} = .8067$$

$$\text{Average } Y_b = \frac{\sum Y_b \text{ at a given power}}{n} = \frac{.2001 + .1899 + .1889}{3} = .1933$$

$$\text{Relative Volatility A} = \frac{Y_a}{X_a} = \frac{.8067}{.0578} = 13.962$$

$$\text{Relative Volatility B} = \frac{Y_b}{X_b} = \frac{.1933}{.9422} = .2052$$

$$\text{Relative Volatility AB} = \sqrt{\alpha_a * \alpha_b} = \sqrt{13.962 * .2052} = 7.021$$

$$\text{Efficiency} = \frac{\text{Theoretical plates}}{\text{Actual number of plates}} * 100 = \frac{7.021}{8} * 100 = 87.758 \%$$

Appendix IV: Individual Team Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4.25	Operating Equipment and completing the Lab
Pre-Lab Editing	1	Compiling the Prelab
Experimental Objective (Prelab)	0.5	Reading lab manual and writing the intro of the lab and the lab goals
Apparatus (prelab)	0.5	Describing and observing the system that will be worked with
Materials & Supplies (prelab)	0.5	Finding and learning how to use the equipment and materials in the lab
Procedure (prelab)	1	Reading and writing how the lab will be completed
Project Safety Evaluation (prelab)	0.25	Assessing all hazardous components, specifically mechanical and electrical
Final Lab Editing	1.5	Making sure the lab report looks nice and professional
Executive Summary (final lab)	1	Brief abstract of the final lab report
Introduction (Final lab)	1	Stating the objecting and including a summary of what will be done in the lab
Theory (Final Lab)	2	Listing and describing equations used to compare theory with experiment
Results (final lab)	9	Showing the results and completing calculations
Discussion (final lab)	3	Explanation of the results. This includes possible errors and how much of an error that there is.
References (final and/or prelab)	0.5	Citing work at the end of the paper and within the paper.
Data Tabulation/Graphs (final)	2	Creating graphs and tables and Transferring them from excel to the final report
Error Analysis (final lab)	1.5	Determining the standard deviation of K and α values
Sample Calculations (final Lab)	3	Showing an example calculation for values that were reported.
Power Point Presentation	1.5	Creation of PowerPoint for presentation.
Total	34	Total time dedicated to this experiment and lab report